

# RAGHAV SAREEN

(650) 660-7750 ♦ Pleasanton, CA

[raghavsareen2006@gmail.com](mailto:raghavsareen2006@gmail.com) ♦ [linkedin.com/in/raghavsareen2006](https://www.linkedin.com/in/raghavsareen2006) ♦ [github.com/RaghavSareen2006](https://github.com/RaghavSareen2006)

## OBJECTIVE

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Computer Engineering student pursuing Firmware and AI Engineering opportunities, with experience spanning embedded C/C++, STM32 motor control, edge-AI computer vision, LLM evaluation, agent testing, Linux systems, and hardware-software integration.

## EDUCATION

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**Bachelor of Science in Computer Engineering**, San Jose State University

Expected May 2028

GPA: 3.776/4.000

*Selected Coursework:* Programming Concepts, Object-Oriented Programming, Algorithms and Data Structures, Assembly Language Programming, Electronics, Circuit Analysis, Introductory Electrical Engineering Lab, Engineering Statistics, Differential Equations and Linear Algebra

## SKILLS

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|---------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Programming</b>              | C, C++, Python                                                                                |
| <b>Firmware &amp; Embedded</b>  | STM32F103, libHAL, GPIO, CAN, PWM, I <sup>2</sup> C, PID control, BLDC motors, limit switches |
| <b>AI &amp; Computer Vision</b> | Ultralytics YOLO, TensorRT, OpenCV, prompt engineering, LLM evaluation, agentic workflows     |
| <b>Systems &amp; DevOps</b>     | NVIDIA Jetson Orin Nano, ARM64 Linux, Docker Compose, Tailscale, Git/GitHub                   |
| <b>AI &amp; Product Tools</b>   | OpenClaw, Claude Code, ChatGPT, Jira, Kanban, Scrum, PRDs, epics, features, user stories      |

## TECHNICAL EXPERIENCE

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**Firmware Team Member, ALICANTO Rover**

Sep 2024 – May 2026

SJSU Robotics

*San Jose, CA*

- Programmed and tested C++ firmware for the ALICANTO rover arm subsystem in preparation for the 2027 University Rover Challenge; committed changes that were reviewed and merged into the shared team codebase.
- Validated three STM32F103 MicroMod and motor assemblies controlling the wrist, shoulder, and elbow, verifying motor response and component behavior with libHAL.
- Updated and tuned PID control functions to reduce arm jitter and improve stability during movement.
- Collaborated with the firmware lead and sublead to implement arm homing for BLDC motors using one limit switch and hands-on GPIO, CAN, and PWM communication.

**AI Agent Testing Volunteer**

Jun 2026

Social Experiments

*Remote*

- Designed 60 functional test cases and executed 37 functional tests plus 13 end-to-end workflow scenarios for an AI Slack agent.
- Tested Jira integration, contract drafting, document ingestion, redlining, approvals, role permissions, duplicate handling, error recovery, and response-time behavior.
- Built a structured test workbook containing priorities, preconditions, prompts, test steps, expected and actual results, automation candidates, and defect notes.
- Identified failures or errors in 28 scenarios, including duplicate responses, long timeouts, incorrect task routing, missing attachments, and incomplete workflow transitions.
- Reported results to the product team, contributing to improvements across most identified issues; used Jira and Agile practices and learned to structure PRDs with epics, features, and user stories.

## PROJECTS

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### AI-Assisted Home Security Arm Turret

Jul 2026 – Present

*NVIDIA Jetson Orin Nano, xArm, YOLO, TensorRT, OpenCV, Docker Compose, Tailscale*

- Conceived and assembled a customizable home-security system by integrating a USB camera and a commercial xArm kit with a Jetson Orin Nano; configured two servo motors for pan and tilt control.
- Used OpenClaw and Claude Code for AI-assisted implementation, then personally deployed, configured, troubleshot, and validated the complete hardware-software system for continuous 24/7 operation at home.
- Tested person and face detection, target verification and reacquisition, automatic room scanning, dog tracking, manual hold, digital zoom, photo capture, Telegram alerts, GPIO status lights, and safe servo limits.
- Deployed GPU inference and web services with Docker Compose and built private remote access through Tailscale; validated an authenticated dashboard with livestream controls and real-time telemetry.

### Jetson Home Lab and Personal AI Infrastructure

Jul 2026 – Present

*Jetson Orin Nano, ARM64 Linux, OpenClaw, Docker Compose, Tailscale, Home Assistant*

- Configured the Jetson as a multi-service home server running OpenClaw, Home Assistant, Music Assistant, Jellyfin, Portainer, and computer-vision containers.
- Managed persistent volumes, host networking, container health, logs, service restarts, and private HTTPS access through Tailscale.
- Diagnosed IoT connection failures, camera conflicts, container crash loops, high system load, swap pressure, and USB HID availability; configured exit-node forwarding and persistent NAT/firewall behavior.
- Created a sanitized Home Assistant configuration repository with Docker Compose files, restore documentation, secret exclusions, and credential-pattern checks.

### The Efficacy of ChatGPT in Mathematics

May 2023 – Sep 2023

*Polygence Research Program*

- Designed and conducted an experiment evaluating ChatGPT across 107 algebra problems from a standardized textbook dataset.
- Measured 81% initial answer accuracy, compared two follow-up strategies, and analyzed failure patterns involving graphs, rational expressions, and mathematical formatting.
- Evaluated LaTeX formatting and math-teacher role prompting as mitigation strategies; authored a research paper and presentation with guidance from a research mentor.
- Won Top Asynchronous Pitch among more than 300 presentations at the Polygence Symposium of Rising Scholars.

## LEADERSHIP & ACTIVITIES

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### Social Media Coordinator, Indian Students Organization

Aug 2025 – Present

Created two promotional reels, participated in four to five event videos, attended weekly planning meetings, and supported promotion and event operations for a program that sold 75 tickets.

### Program Member, MESA Engineering Program

Sep 2024 – Present

Volunteered 12+ hours as an event floater for Science Extravaganza, supporting hands-on STEM workshops serving more than 350 middle-school students.

## AWARDS & CERTIFICATIONS

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**Awards:** 2026 President's Scholar, San Jose State University; Top Asynchronous Pitch, Polygence Symposium of Rising Scholars; AP Scholar; California State Seal of Biliteracy

**Certifications:** Command Line in Linux and Getting Started with Linux Terminal (Coursera, 2026); Learn Git and GitHub in One Day (Udemy, 2026); Claude Code in Action (Anthropic, 2026)

## LANGUAGES

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English (Full Professional Proficiency) ◇ Hindi (Native or Bilingual Proficiency) ◇ Spanish (Limited Working Proficiency)